Signal Rescue Lab

Hamming(7,4) student worksheet | Grades 9-10 | Even parity

Name: ____________________	Team: ____________________	Date: ____________________

Finish line: independently encode four message bits into seven, repair any single flipped bit, and recover the original message.

1A The impossible repair

A transmission arrives as: 1 0 1 ? 1 1 0

What was the missing bit? Explain why the receiver can or cannot know.

1B Even-parity sprint

Add one check bit so the total number of 1s is even.

data	check bit	total 1s	audit: even?
101			
111			
000			
0101			

1C Address detective

Fill YES when a position belongs to a check. Then write the failed-label sum for positions 3, 5, 6, and 7.

position	1	2	3	4	5	6	7
check 1							
check 2							
check 4							

Addresses: 3 = ____________ 5 = ____________ 6 = ____________ 7 = ____________

Build the seven-slot machine

Page 2 | Position roles, check groups, and a three-message-bit warm-up

2A Label the slots

Write check 1, check 2, data 1, check 4, data 2, data 3, data 4 under positions 1-7.

1	2	3	4	5	6	7

Reference: check 1 covers 1,3,5,7 | check 2 covers 2,3,6,7 | check 4 covers 4,5,6,7

2B Training wheels: encode 1 0 1

Place 1,0,1 in positions 3,5,6. Fix position 7 at 0. Choose each check bit to make its group even.

1	2	3	4	5	6	7
		1		0	1	0
check 1	check 2	data 1	check 4	data 2	data 3	fixed 0

check 1 count: __________ even? ☐ | check 2 count: __________ even? ☐ | check 4 count: __________ even? ☐

Completed seven-bit word: ____

2C One more restricted warm-up

Encode message bits 0 1 1 with position 7 fixed at 0.

1	2	3	4	5	6	7
		0		1	1	0
check 1	check 2	data 1	check 4	data 2	data 3	fixed 0

Completed seven-bit word: ____

Accuracy note: this is a restricted practice version of the full Hamming(7,4) code, not the standard Hamming(7,3) code.

Encode Hamming(7,4)

Page 3 | Four message bits become seven transmitted bits

3A Write the four-move routine

1. Label	2. Place data	3. Set checks	4. Verify

3B Encode the message 1 0 1 1

1	2	3	4	5	6	7
		1		0	1	1
check 1	check 2	data 1	check 4	data 2	data 3	data 4

check 1 count: ___________ even? ☐	check 2 count: ___________ even? ☐	check 4 count: ___________ even? ☐

Codeword: ________________________________ Audit complete? ☐

3C Speed round

Encode each message. A teammate should audit all three checks.

message	seven-bit codeword	audit initials	all even?
0000			☐
1111			☐
0101			☐
1100			☐

Encoder shortcut: data always goes in positions 3,5,6,7. Check bits always go in 1,2,4.

Find and repair one bad bit

Page 4 | Run checks, add failed labels, flip once, extract data

4A Repair 0 1 1 0 0 0 1

1	2	3	4	5	6	7
0	1	1	0	0	0	1
check 1	check 2	data 1	check 4	data 2	data 3	data 4

check	positions	number of 1s	pass/fail	failed label
1	1,3,5,7			
2	2,3,6,7			
4	4,5,6,7			

Syndrome address: _______ Flip position: _______ Corrected word: ____________________

Recovered message from positions 3,5,6,7: ____________________

4B Packet repair race

packet	received	error position	corrected / message
A	0101101		
B	0111101		

Decoder routine: 1 run checks -> 2 add failed labels -> 3 flip that position -> 4 extract 3,5,6,7

Partner packet exchange

Page 5 | Create, corrupt, swap, rescue

5A Sender lab

Choose any four-bit message, encode it, have your partner audit it, then flip exactly one bit.

Message: ____________________ Valid codeword: ____________________

1	2	3	4	5	6	7
check 1	check 2	data 1	check 4	data 2	data 3	data 4

Secret flipped position: _______ Corrupted transmission to send: ____________________

5B Receiver rescue

Received transmission: ____________________

check	count of 1s	pass/fail	failed label
1			
2			
4			

Syndrome: _______ Corrected codeword: ____________________

Recovered message: ____________________ Sender confirms? ☐

Success means the corrected codeword, recovered message, and explanation all agree.

Solo certification

Page 6 | Notes allowed; no partner; show every check

6A Part A - encode 1 0 0 1

1	2	3	4	5	6	7
		1		0	0	1
check 1	check 2	data 1	check 4	data 2	data 3	data 4

check 1 count: __________ even? ☐	check 2 count: __________ even? ☐	check 4 count: __________ even? ☐

Final codeword: ____

6A Part B - repair 0 0 1 1 1 0 1

1	2	3	4	5	6	7
0	0	1	1	1	0	1
check 1	check 2	data 1	check 4	data 2	data 3	data 4

Failed checks: ___________________ Error position: ______

Corrected codeword: ___________________ Recovered message: ___________________

6B Bonus - extend to Hamming(8,4)

Add one overall even-parity bit as position 8. Complete the decision table.

Hamming syndrome	overall parity	what should the receiver do?
0	even	
nonzero	odd	
0	odd	
nonzero	even	

Exit reflection: The step I am least likely to forget is ________________________________ because

____